CSE STUDY HUB

Generative AI & Prompt Engineering

UNIT 4 · Applications of Generative AI Across Domains

What this unit is about

This unit builds understanding of **Applications of Generative AI Across Domains**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. Content creation, advertising and digital marketing

2. Education: tutoring, lesson generation and personalised learning

3. Healthcare: documentation and decision support with professional oversight

4. Finance and business: summarisation, analysis and customer support

5. Software development: code generation, debugging and documentation

6. Research and productivity: ideation, synthesis and workflow automation

7. Domain risks: privacy, bias, accuracy, regulation and human review

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 4 NOTES · PART 1

Content creation, advertising and digital marketing

Content creation, advertising and digital marketing is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Education: tutoring, lesson generation and personalised learning

Education: tutoring, lesson generation and personalised learning is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Healthcare: documentation and decision support with professional oversight

Healthcare: documentation and decision support with professional oversight is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Applications of Generative AI Across Domains, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 4 NOTES · PART 2

Finance and business: summarisation, analysis and customer support

Finance and business: summarisation, analysis and customer support is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Software development: code generation, debugging and documentation

Software development: code generation, debugging and documentation is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Applications of Generative AI Across Domains, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 4 NOTES · PART 3

Research and productivity: ideation, synthesis and workflow automation

Present Research and productivity: ideation, synthesis and workflow automation as an ordered workflow. State the preconditions, list the major steps, explain what each step produces, and mention how errors are detected or handled. Use a small worked example or flowchart in the exam.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Domain risks: privacy, bias, accuracy, regulation and human review

For Domain risks: privacy, bias, accuracy, regulation and human review, discuss the objective, stakeholders and possible harm. Cover consent/authority, confidentiality, integrity, accountability, transparency and applicable policy or law. Recommend safeguards and explain how they are checked.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Applications of Generative AI Across Domains, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **Content creation, advertising and digital marketing** — know the definition, core working and one example.

• **Education: tutoring, lesson generation and personalised learning** — know the definition, core working and one example.

• **Healthcare: documentation and decision support with professional oversight** — know the definition, core working and one example.

• **Finance and business: summarisation, analysis and customer support** — know the definition, core working and one example.

• **Software development: code generation, debugging and documentation** — know the definition, core working and one example.

• **Research and productivity: ideation, synthesis and workflow automation** — know the definition, core working and one example.

• **Domain risks: privacy, bias, accuracy, regulation and human review** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from Applications of Generative AI Across Domains. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define Content creation, advertising and digital marketing and explain its importance.

Q2. Explain the working/steps involved in Education: tutoring, lesson generation and personalised learning.

Q3. Differentiate between two related concepts from Applications of Generative AI Across Domains.

Q4. Give one practical application of Domain risks: privacy, bias, accuracy, regulation and human review and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of Applications of Generative AI Across Domains.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.